

EX 207 (a)

IPC (shared memory)

Sender

Aim:-

To implement inter process communication using shared memory in C. Where one process writes data into a shared memory.

Code :-

```
#include <stdio.h>
#include <sys/types.h>
#include <system.h>
#include <string.h>
```

```
in main () {
```

```
key_t key = 1234;
```

```
int shmidx;
```

```
shmidx = shmget (key, 1024, 0666 | IPC_CREAT);
```

```
char *str = (char *) shmat (shmidx, (void *) 0, 0);
```

```
printf ("Enter message");
```

```
scanf ("%s", str);
```

```
printf ("Data written to shared memory:
```

```
%s", str);
```

```
return 0;
```

Output:

Enter msg: Hello

Data written to shared memory: Hello

Result:-

Thus the program successfully creates a shared memory segment and writes. Thus input on it.

```

Code:
#include <stdio.h>
#include <stdlib.h>
#include <sys/types.h>
#include <unistd.h>

int main()
{
    int *ptr;
    ptr = (int *) malloc(1024);
    if (ptr == NULL)
    {
        printf("Error\n");
        return 1;
    }
    printf("Enter message\n");
    char *str = (char *) ptr;
    fgets(str, 1024, stdin);
    printf("Data written to shared memory\n");
    return 0;
}

```

Output:
 Enter message: Hello
 Data written to shared memory: Hello
 Thus the program successfully creates a shared memory

Ex: 7(b)

Aim: To read data from a shared memory segment using IPC where ~~segment~~ requires access and display message.

Code:-

```
#include <stdio.h>
#include <sys/types.h>
#include <sys/shm.h>

int main() {
    key_t key = 1234;
    int shmid;
    shmid = shmget(key, 1024, 0666);
    char *str = (char *) shmat(shmid, (void *) 0, 0);
    printf("Enter message:");
    scanf("%s", str);
    printf("Data written to shared memory: %s\n", str);
    return 0;
}
```

Output:

Enter msg: Hello

Data written to shared memory: Hello

Result:

Thus the program successfully create a shared memory